Technical Document #590: Software Engineering - Comprehensive Overview

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API design defines how software components communicate. RESTful APIs and GraphQL are popular API design approaches.

Code review examines code changes before merging. It improves code quality, catches bugs, and shares knowledge across the team.

Continuous deployment (CD) automatically deploys code changes to production. It enables rapid, reliable releases.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Domain-driven design (DDD) models software around business domains.
- Microservices architecture structures applications as independently deployable services.
- API design defines how software components communicate.